

Second Virial Cross-Coefficients: Correlation and Prediction of k_{ij}

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The prediction of the second virial cross-coefficient B_{ij} depends most sensitively on the mixing rule for the characteristic critical constant T_{cij} (Equation 5). The characteristic binary constant, k_{ij} , can be determined by fitting B_{ij} and other data. The prediction of B_{ij} , therefore, requires knowledge of k_{ij} . The k_{ij} values are available for binaries of inorganic gases and hydrocarbons with up to 8 carbon atoms. Extensions in the data base and correlations for k_{ij} 's are presented here in two areas. First, recent B_{ij} data for binaries involving C_{10} – C_{30} hydrocarbons are reduced to k_{ij} 's, which are shown to be both meaningful and satisfactory in the calculation of B_{ij} over the entire carbon-number range. A similar approach is taken in the second area investigated, binaries involving polar compounds. New k_{ij} data are used to establish trends and develop preliminary k_{ij} correlations so that B_{ij} and, more importantly, fugacity coefficients can be predicted reliably.

In the analysis and correlation of vapor–liquid equilibrium (VLE) data it is essential, especially at superatmospheric pressures, to take into account the effect of vapor-phase nonideality. This is expressed by the fugacity coefficient which, as long as the density of the mixture is not greater than one fourth of its critical value, can be calculated reliably with the following equation (for a binary mixture):

$$\ln \phi_i = \frac{2}{V_m} (y_i B_{ii} + y_j B_{ij}) - \ln z_m \quad (1)$$

where ϕ_i , y_i , and B_{ii} are the fugacity coefficient, vapor mole fraction, and second virial coefficient of Component i , while v_m and z_m are the molar

volume and compressibility factor of the vapor mixture. It is the prediction of B_{ij} , the second virial cross-coefficient, that will primarily concern us.

B_{ij} will be predicted with an empirical correlation that was proposed in 1974 (1). The original correlation was developed primarily for oxygenated polar compounds, but it has been extended to haloalkanes (2) and water pollutants (3). Refs. 1, 2, and 3 can be consulted for details on the development of the correlation and the results.

Nonpolar Mixtures

For nonpolar compounds and mixtures, the correlation is a modified form of the Pitzer–Curl relationship (*see* Ref. 1):

$$\frac{BP_c}{RT_c} = f^{(0)}(T_R) + \omega f^{(1)}(T_R) \quad (2)$$

$$f^{(0)}(T_R) = 0.1445 - 0.330/T_R - 0.1385/T_R^2 - 0.0121/T_R^3 - 0.000607/T_R^8 \quad (3)$$

$$f^{(1)}(T_R) = 0.0637 + 0.331/T_R^2 - 0.423/T_R^3 - 0.008/T_R^8 \quad (4)$$

The same equations are used for B_{ij} , but with characteristic parameters $P_{c_{ij}}$, $T_{c_{ij}}$, and ω_{ij} in place of the pure-component constants. The former are related to the latter through the following mixing rules:

$$T_{c_{ij}} = (T_{c_i} T_{c_j})^{1/2} (1 - k_{ij}) \quad (5)$$

$$P_{c_{ij}} = \frac{4 T_{c_{ij}} (P_{c_i} V_{c_i}/T_{c_i} + P_{c_j} V_{c_j}/T_{c_j})}{(V_{c_i}^{1/3} + V_{c_j}^{1/3})^3} \quad (6)$$

$$\omega_{ij} = 0.5(\omega_i + \omega_j) \quad (7)$$

The most sensitive mixing rule is Equation 5. A characteristic constant for each binary, k_{ij} , expresses the deviation from the geometric mean for $T_{c_{ij}}$, which applies (that is, $k_{ij} = 0$) only when i and j are very similar in size and chemical nature. Otherwise, in the absence of any strong specific chemical interaction between i and j , k_{ij} should be positive and thus $T_{c_{ij}}$ would be less than the geometric mean.

An extensive tabulation of k_{ij} 's for nonpolar systems was reported by Chueh and Prausnitz in 1967 (4) and, with a few additions, in 1968 (5). The k_{ij} 's presented by Chueh and Prausnitz are primarily for binaries of hydrocarbons with up to 8 carbon atoms—along with a few naphthalene systems. There are also data for binaries of CO_2 , H_2S , N_2 , Ar, etc. Most

of them were obtained from experimental information on the second virial coefficient or the saturated liquid volume of binary systems, although the resulting k_{ij} 's then were used by Chueh and Prausnitz in their calculations with a modified form of the Redlich-Kwong (RK) equation of state. What this says (and it is an important point) is that once the value of k_{ij} is determined for an i/j binary, by fitting (reliable) B_{ij} or other mixture data, it can be used in the prediction of any other mixture property. That is, under ideal conditions, k_{ij} is a true constant that can be used with any equation of state or correlation that uses $T_{c_{ij}}$.

Another benefit drawn from using k_{ij} to correct for the deviation of $T_{c_{ij}}$ from the geometric mean is that it also can be used to predict ϵ_{ij} , the characteristic energy parameter for a binary molecular interaction. Since ϵ is proportional to T_c , it follows that

$$\epsilon_{ij} = (\epsilon_i \epsilon_j)^{1/2} (1 - k_{ij}) \quad (8)$$

The more theoretically oriented investigators probably consider the k_{ij} in Equation 8 to be physically more significant than that in Equation 5. However, the two should be very similar numerically—especially if we consider that the uncertainty in the determination of k_{ij} is at least ± 0.01 units.

Hiza and Duncan (6) have examined the deviation of ϵ_{ij} from the geometric mean primarily for cryogenic systems. They also developed a correlation for the prediction of k_{ij} for such systems that will be discussed later.

The k_{ij} tabulations of Hiza and Duncan and of Chueh and Prausnitz are adequate for nonpolar systems involving inorganic gases and hydrocarbons with perhaps up to 8 carbon atoms. However, for process calculations on heavy hydrocarbons, coal liquids, etc., k_{ij} must be known or predicted for binaries of hydrocarbons with carbon number (much) greater than 10.

Available B_{ij} data on binaries involving hydrocarbons with up to 30 carbon atoms were fitted with Equations 2–7 to determine optimum values for the characteristic constant k_{ij} . These results, which will be examined below, have also helped establish trends and even suggest ways of correlating and predicting k_{ij} 's for hydrocarbon binaries. Before the new information is reviewed, therefore, a few comments are appropriate on what is known and what has been done in attempting to correlate k_{ij} with pure-component properties.

A Review of Correlations for k_{ij} . Justification for using the geometric mean for ϵ_{ij} —and for $T_{c_{ij}}$ —is provided by London's theory of dispersion forces. When London's expression for the intermolecular energy is equated to the attractive part of the Lennard-Jones (LJ) 12:6 potential, the following relationship obtains:

$$k_{ij} = 1 - \left[\frac{2 (I_i I_j)^{1/2}}{I_i + I_j} \right] \left[\frac{2 (\sigma_i \sigma_j)^{1/2}}{\sigma_i + \sigma_j} \right]^6 \quad (9a)$$

where I_i is the first ionization potential and σ_i is the collision diameter of i . If the latter is replaced by the critical volume, Equation 9a assumes the form

$$k_{ij} = 1 - \left[\frac{2 (I_i I_j)^{1/2}}{I_i + I_j} \right] \left[\frac{2 (V_{ci} V_{cj})^{1/6}}{V_{ci}^{1/3} + V_{cj}^{1/3}} \right]^6 \quad (9b)$$

Equation 9 may have some theoretical basis, but at best it provides only rough estimates for the k_{ij} parameter. Indeed, better results are sometimes obtained when the exponent of the second bracketed quantity is changed to 3. For paraffin-paraffin mixtures only, Chueh and Prausnitz (7) found that

$$k_{ij} = 1 - \left[\frac{2 (V_{ci} V_{cj})^{1/6}}{V_{ci}^{1/3} + V_{cj}^{1/3}} \right]^3 \quad (10)$$

gives satisfactory results. (Even for the methane-*n*-decane binary, the first bracketed quantity in Equation 9 is equal to 0.992 and hence can be ignored.) As Teja (8) has shown, Equation 10 follows from Berthelot's mixing rule for the van der Waals (VDW) a_{ij} parameter: $a_{ij} = (a_i a_j)^{1/2}$.

The inadequacy of Equation 9 may result from the fact that its derivation ignored the contribution of the repulsive forces. Taking these forces into account, however, would make the theoretical treatment of the problem formidable, and therefore Hiza and Duncan (6) opted for an empirical approach. Their analysis of data for binaries of H₂, He, and Ne with light hydrocarbons and Ar suggested the use of I as the correlating parameter for k_{ij} in the following form:

$$k_{ij} = 0.17 (I_i - I_j)^{1/2} \ln \frac{I_i}{I_j} \quad (11)$$

where i is the component with the larger ionization potential. With the exception of the O₂ binaries, Equation 11 satisfies binaries of inorganic gases, methane, ethane, and ethylene. Agreement with experimental values was within ± 0.04 . More recently, it was shown by H.-M. Lin that Equation 11 (with 0.18 in place of 0.17) gives a good representation of the data for rare-gas mixtures (9).

Equations 10 (for paraffin-paraffin mixtures) and 11 (for mixtures of inorganic gases, CH₄, C₂H₆, and C₂H₄) are probably the only reasonably reliable correlations available in the literature for the prediction of k_{ij} for nonpolar mixtures.

Finally, Fender and Halsey (10) proposed a geometric–arithmetic mean that follows, by drastic simplification, from the Kirkwood–Müller formula for ϵ_{ij} :

$$\epsilon_{ij} \cong \frac{2 \epsilon_i \epsilon_j}{\epsilon_i + \epsilon_j}$$

This relationship, which gives very good results for Ar–Kr, leads to the following expression for k_{ij} :

$$k_{ij} = 1 - \frac{2 (\epsilon_i \epsilon_j)^{1/2}}{\epsilon_i + \epsilon_j} \quad (12a)$$

$$= 1 - \frac{2 (T_{c_i} T_{c_j})^{1/2}}{T_{c_i} + T_{c_j}} \quad (12b)$$

Equation 12 will be examined in the next section.

The k_{ij} 's for Methane–Hydrocarbon Binaries. As noted earlier, most of the k_{ij} values available in the literature are for binaries of inorganic gases and hydrocarbons with up to 8 carbon atoms. New determinations of optimum k_{ij} values for methane–hydrocarbon binaries are listed in Table I. These and Chueh's results make a total of 26 k_{ij} values (for 25 binaries; two different k_{ij} 's were included for the naphthalene binary) over the carbon-number range of 2 to 30 for j . This is the most

Table I. Optimum k_{ij} Values for Methane–Hydrocarbon Binaries

j	Average Deviation of B_{ij} (cm^3/gmol)	k_{ij}	t Range ($^{\circ}\text{C}$) (No. Points)	Refer- ences
Benzene	0.8	0.09	50 (1)	11
2,2,5-Trimethylhexane	3	0.19	25–100 (4)	12
<i>n</i> -Decane	13	0.16	50–125 (4)	12
<i>tert</i> -Butylbenzene	6	0.21	50–125 (4)	12
Naphthalene	12	0.13	21– 68 (6)	13
1-Methylnaphthalene	6	0.24	75–175 (3)	14
<i>n</i> -Dodecane	4	0.15	75–150 (4)	12
Bicyclohexyl	27	0.20	50–170 (4)	14
Diphenylmethane	17	0.12	65–170 (4)	14
Phenanthrene	11	0.18	40–138 (7)	15
Anthracene	8	0.19	66–185 (7)	16
<i>n</i> -Hexadecane	21	0.22	75–175 (5)	14
<i>n</i> -Eicosane	8	0.21	165–270 (4)	14
Squalane ^a	22–25	0.33 $\pm$ 0.02	230, 272 (2)	14

^a 2,6,10,15,19,23-Hexamethyltetracosane.

extensive set of k_{ij} 's available for a given Component i , and therefore was selected to check the various relationships presented in the previous section for correlating k_{ij} .

As expected, Equation 10 worked well for the 14 paraffin binaries; the average deviation of the predicted k_{ij} from the optimum values was only 0.02. However, when all 26 hydrocarbon binaries were included, the average deviation increased to 0.05.

Equation 11 was much less satisfactory, even when the coefficient was adjusted to minimize the deviation. The optimum value of the coefficient was 0.18, in agreement with Lin's conclusion for rare-gas mixtures, but the root-mean-square deviation of k_{ij} was 0.08. Any relationship involving I , the first ionization potential, is doomed to failure when applied to heavy hydrocarbon mixtures because I , for a given homologous series, is very weakly dependent on carbon number. For example, the I of n -decane is 10.19 eV, only 0.24 eV less than that of n -hexane, while that of n -eicosane should be about 10.04 eV. Thus, a correlation of k_{ij} 's for methane-paraffin binaries based solely on ionization potentials would give the same result for all $C_{10}+$ paraffins.

Equation 12b works surprisingly well. Although it is inferior to Equation 11 when applied to the paraffin binaries (average deviation of 0.03 vs. 0.02), it comes out better when all hydrocarbon binaries are considered (average deviation of 0.04 vs. 0.05). Equation 12b is also satisfactory for the binaries of methane with Ar, Kr, N_2 , and H_2S (average deviation of 0.01 vs. 0.03 with Equation 10), but is very poor for methane- H_2 .

Both the critical volume and the critical temperature are therefore satisfactory correlating parameters for k_{ij} . However, the best correlating parameter for all 26 methane-hydrocarbon k_{ij} values was the carbon number of the hydrocarbon. This is shown in Figure 1, where all the data have been plotted along with the simple relationship

$$k_{ij} = 0.0279 [\ln (n_{c_j})]^2 \quad (13)$$

where n_{c_j} is the carbon number of the hydrocarbon. The average deviation of the k_{ij} predicted with Equation 13 is only 0.02—for all 26 points. (The largest deviations from the correlation in Figure 1 are for both paraffins and aromatics.) Thus, Equation 13 provides the best correlation for the effect of the hydrocarbon carbon number on k_{ij} . The same approach will be used now with other binaries for which extensive data are available.

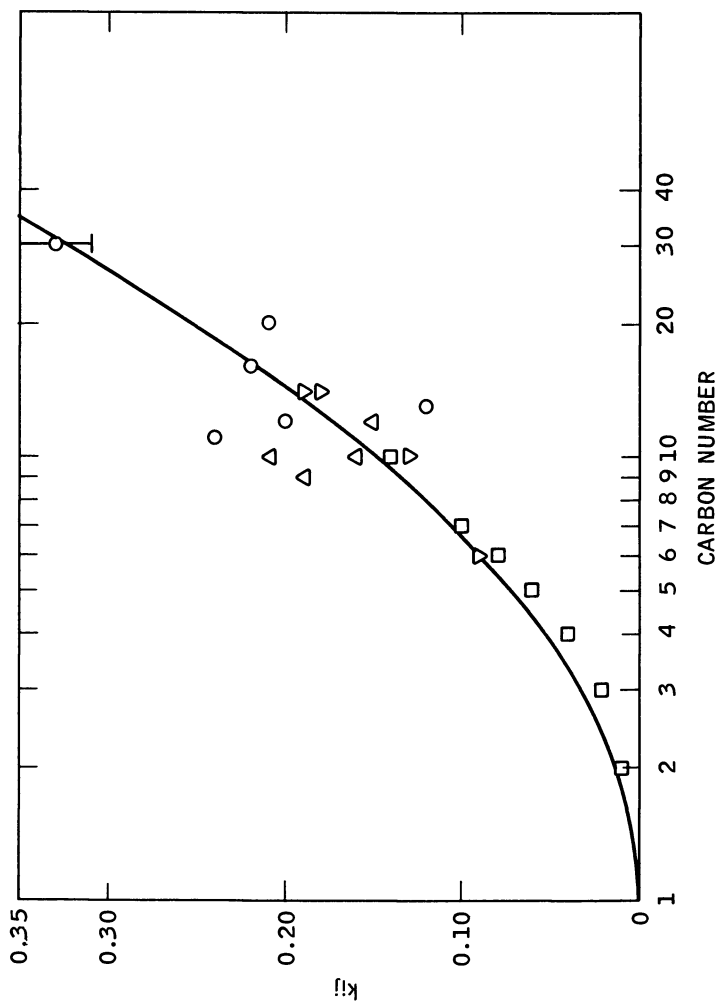


Figure 1. Optimum k_{ij} values for methane-hydrocarbon binaries: (—), Equation 13; (○) Kaul and Prausnitz (1978); (Δ), D'Avila et al. (1976); (□), results of Chueh and Prausnitz (1967); (▽), King et al. (1966-1970).

The k_{ij} 's for Other Hydrocarbon Binaries. Table II presents optimum k_{ij} values for heavy hydrocarbon binaries of ethylene and ethane. The two sets of k_{ij} 's are very similar, and this is shown more clearly in Figure 2, where the results of Chueh and Prausnitz (4) have been plotted also. Chueh and Prausnitz actually found that there was no significant difference in the k_{ij} 's between ethylene and ethane—or between *n*-hexane and benzene.

The line in Figure 2 was calculated with a more general form of Equation 13:

$$k_{ij} = m[\ln(n_{c_j} - n_{c_i} + 1)]^2 \quad (14)$$

(This form ensures that $k_{ij} = 0$ when $n_{c_j} = n_{c_i}$; it is implied that $n_{c_j} \geq n_{c_i}$.) For the ethylene and ethane binaries, the value of m is 0.0202. A substantially better fit of the data is possible if k_{ij} is assumed to be linearly dependent on carbon number:

$$k_{ij} = 0.0117(n_{c_j} - n_{c_i} - 1); \quad n_{c_j} > n_{c_i} \quad (15)$$

Teja (8) recommended Equation 15 (with 0.01 as the coefficient) for the paraffin binaries of ethane.

Table II. Optimum k_{ij} Values for Ethylene and Ethane Binaries

<i>i</i>	<i>j</i>	Average Deviation of B_{ij} (cm^3/gmol)	k_{ij}	<i>t</i> Range ($^{\circ}\text{C}$) (No. Points)	Refer- ences
<i>i</i> = Ethylene					
	Naphthalene	8	0.09	23–69 (9)	13, 17
	1-Methylnaphthalene	37	0.10	75–175 (3)	14
	Bicyclohexyl	19	0.14	50–170 (3)	14
	Diphenylmethane	21	0.12	65–175 (3)	14
	Phenanthrene	26	0.15	37–142 (7)	15
	Anthracene	11	0.15	65–180 (7)	16
	<i>n</i> -Hexadecane	9	0.15	75–175 (3)	14
<i>i</i> = Ethane					
	Naphthalene	8	0.12	26– 67 (7)	18
	1-Methylnaphthalene	24	0.08	75–175 (3)	14
	Bicyclohexyl	23	0.10	50–170 (3)	14
	Diphenylmethane	18	0.12	65–175 (3)	14
	Anthracene	8	0.16	63–175 (6)	16
	<i>n</i> -Hexadecane	18	0.12	75–175 (3)	14
	<i>n</i> -Eicosane	6	0.19	165–270 (4)	14
	Squalane	21	0.29	230, 272 (2)	14

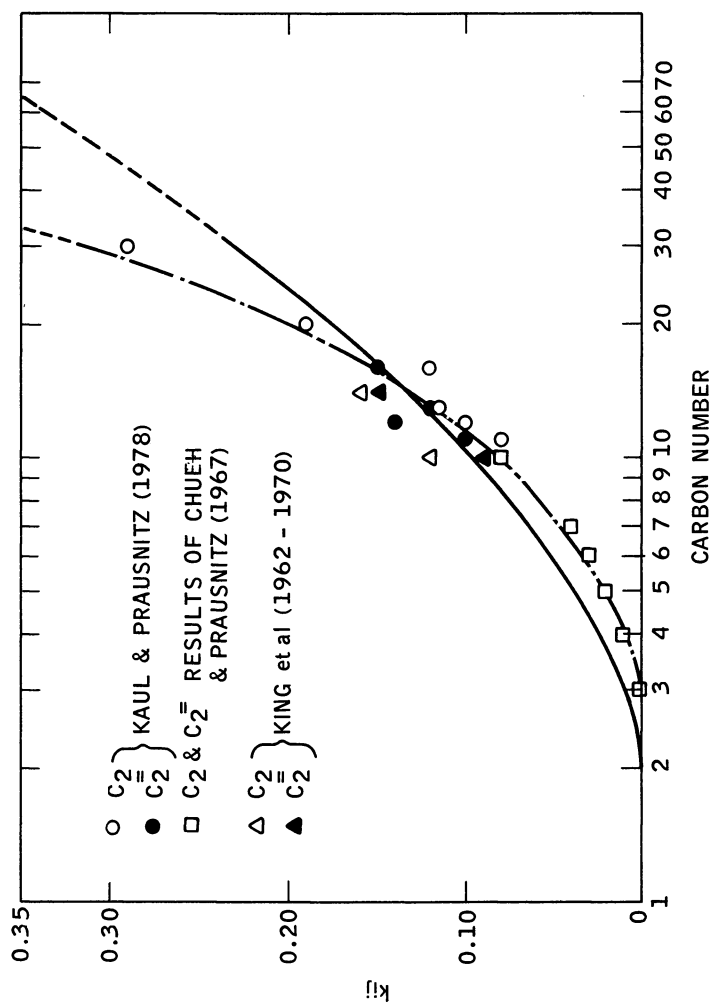


Figure 2. Optimum k_{ij} values for ethylene- and ethane-hydrocarbon binaries: (— · —), Equation 15; (—), Equation 14 with $m = 0.0202$.

The similarity between ethylene and ethane or even benzene and *n*-hexane no longer holds when some specific chemical interaction takes place between *i* and *j*. A good example of this is provided by the acetylene binaries, which have substantially higher k_{ij} 's than the ethylene or ethane binaries. Furthermore, the k_{ij} for acetylene-ethylene is 0.02 units lower than that for acetylene-ethane (4).

The k_{ij} values for N₂-hydrocarbon binaries are listed in Table III and plotted on Figure 3—along with the results of Chueh and Prausnitz (4). As shown in Figure 3, Equation 14 (with $m = 0.0364$) gives a satisfactory fit of the data. That is, Equation 14, with $n_{ci} = 0$, applies to inorganic-hydrocarbon binaries as well. Indeed, it appears to fit reasonably well the k_{ij} 's for the hydrocarbon binaries of H₂—the most important inorganic gas in the heavy hydrocarbon and synthetic fuels processes—and even of Ar. The results of new k_{ij} determinations for the H₂ and Ar binaries are included in Table III, and all the values have been plotted on Figure 4.

The similarity of the k_{ij} 's for the binaries of N₂, H₂, and Ar naturally suggests that the three must have similar values for a property that

Table III. Optimum k_{ij} Values for Inorganic-Hydrocarbon Binaries

<i>j</i>	Average Deviation of B_{ij} (cm^3/gmol)	k_{ij}	<i>t</i> Range (°C) (No. Points)	Refer- ences
<i>i</i> = N ₂				
Benzene	2	0.135 ± 0.015	35– 50 (5)	11 ^a
2,2,5-Trimethyl- hexane	2	0.24	25–100 (4)	12
<i>n</i> -Decane	5	0.18	50–125 (4)	12
<i>tert</i> -Butylbenzene	3	0.25	50–125 (4)	12
Naphthalene	3	0.19	22, 72 (2)	17
<i>n</i> -Dodecane	4	0.20	75–150 (4)	12
<i>i</i> = H ₂				
Benzene	—	0.175 ± 0.045	50 (2)	11 ^a
Naphthalene	4	0.18	22, 70 (2)	17
<i>i</i> = Ar				
Benzene	1	0.21	25 (1)	19
Benzene	6	0.14	22, 50 (2)	11
Naphthalene	7	0.23	24–74 (3)	17, 19
Anthracene	1	0.25	75 (1)	19

^a Optimum k_{ij} value also reflects data of Gainey and Young (N₂) and of Everett et al. (N₂, H₂) (11).

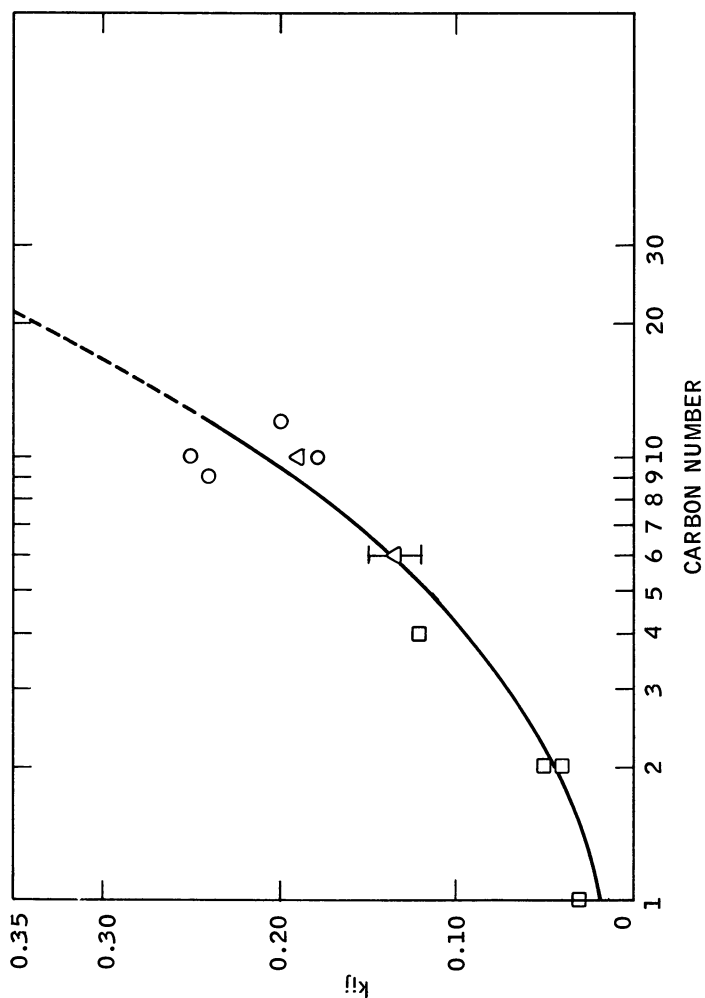


Figure 3. Optimum k_{ij} values for N_2 -hydrocarbon binaries: (—), Equation 14 with $m = 0.0364$; (O), D'Avila et al. (1976); (□), results of Chueh and Prausnitz (1967); (Δ), King et al. (1962, 1969).

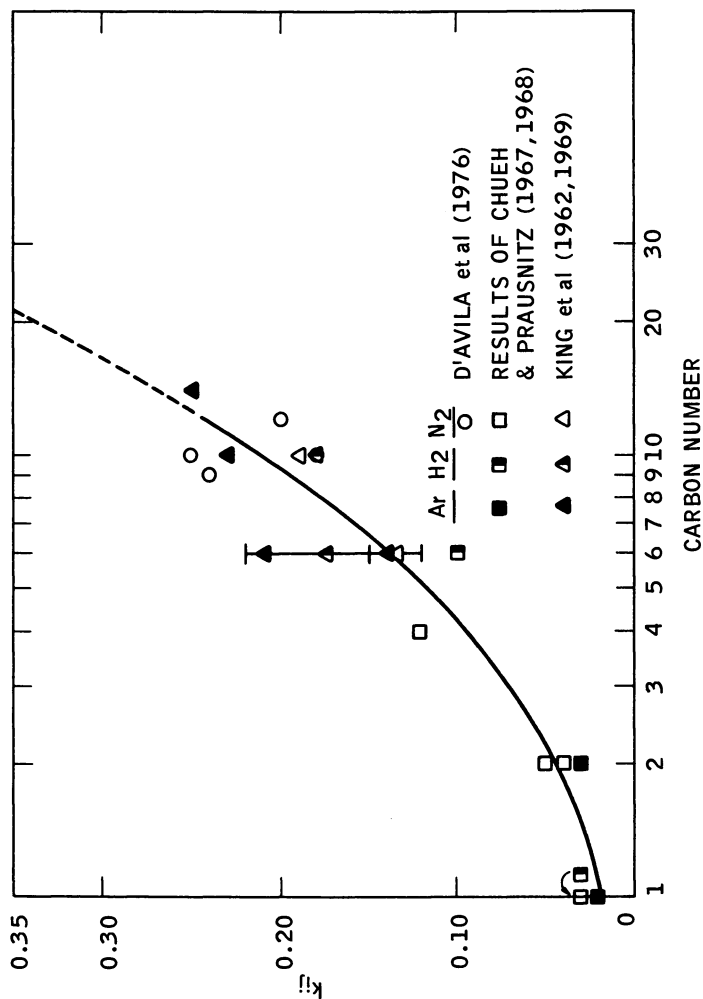


Figure 4. Optimum k_{ij} values for N_2 , H_2 , and Ar-hydrocarbon binaries: (—), Equation 14 with $m = 0.0364$.

affects k_{ij} . Their critical volumes are only roughly similar (51.5, is the "classical" constant for H_2 used in B_{ij} calculations (3)):

	V_c ($cm^3/gmol$)
N_2	89
H_2	51.5
Ar	75

but their first ionization potentials are nearly exactly equal (20)

	I , eV
N_2	15.5
H_2	15.4
Ar	15.4

The first ionization potential may prove useful in sorting out and ordering the data for binaries of inorganic compounds, but it is certainly of little use in the prediction of k_{ij} 's for hydrocarbon-hydrocarbon binaries. For example, the I of n -pentane (10.55 eV) is very similar to that of ethylene (10.51 eV), but the k_{ij} 's for the n -pentane binaries are markedly lower than those for the corresponding ethylene binaries (4). On the other hand, the ethylene and ethane binaries have similar k_{ij} 's (Figure 2), even though the I of ethane is 1.25 eV higher than for ethylene (20).

Concluding Remarks on Nonpolar Mixtures. Plotting k_{ij} for a given Component i against the carbon number of the hydrocarbon j provides a convenient way for checking individual values and trends. However, this should not be taken to imply that the carbon number is a good correlating parameter for all cases. For hydrocarbon binaries of components not considered here (or in Refs. 4, 5, and 6), the k_{ij} still may be predicted with either Equation 10 (if both components are paraffins) or 12b (for all others). However, if m can be correlated with some pure-component property (perhaps V_c , although I works better for inorganics), Equation 14 probably will prove superior to Equations 10 and 12b for all hydrocarbon binaries.

Actually, Equation 15 may prove to be the best relationship for hydrocarbon-hydrocarbon binaries with $n_{c_i} \geq 2$ (and $n_{c_j} > n_{c_i}$). It certainly works very well for $n_{c_i} = 2$, but there are insufficient data for $n_{c_i} > 2$ to determine the optimum form of the relationship and the value(s) of the coefficient.

Figures 1-4 show that the extension of the k_{ij} approach to heavy hydrocarbons is reasonable and more or less straightforward. For the moderately dense vapor mixtures considered here, there is continuity between low- and high-molecular-weight systems.

Although k_{ij} apparently retains its usefulness up to very high carbon numbers, the description of the properties of liquid mixtures of $C_{10}+$ hydrocarbons is not completely satisfactory when only k_{ij} is used. An additional parameter is required, most likely as a correction for the deviation of σ_{ij} from the arithmetic mean (or $V_{c,ij}$ from the Lorentz or cube-root mean). Such a correction has been used even for cryogenic mixtures (9), but it should be more meaningful—and important—for mixtures of heavy hydrocarbons. If the value of k_{ij} has been fixed already by fitting B_{ij} or some other property of the vapor mixture, then the analysis of VLE data will permit the unique determination of the second parameter.

Polar Mixtures

The polar mixtures should be divided into polar-nonpolar and polar-polar systems. Very little can be said about the latter because the data are limited and the polar-polar interactions are generally too system-specific to allow any generalizations. Certainly, the various correlations already examined for the nonpolar mixtures cannot be expected to work for polar-polar binaries. Indeed, they are unsatisfactory even for most polar-nonpolar binaries.

The k_{ij} 's for a few polar-polar binaries have been presented in Refs. 1 and 2. In addition, Table VI of Ref. 1 presents average k_{ij} values that can be used, at least as a rough guide, for mixtures comprised of ketones, ethers, alcohols, and water.

Polar-Nonpolar Binaries. The B_{ij} for polar-nonpolar binaries is assumed to have no polar term (1), and therefore the calculation is carried out exactly as for nonpolar binaries; that is, with Equations 2-7.

Average values for the k_{ij} of polar-hydrocarbon binaries were given in 1974, in the first paper on the new correlation (1). They were as follows

i (j = hydrocarbon)	k_{ij}
ketones	0.13
ethers	0.10
alcohols	0.15
water	0.40

In addition, the data analysis in Ref. 2 suggested that $k_{ij} \cong 0.05$ for haloalkane-hydrocarbon binaries.

New B_{ij} data on binaries of ethanol (21), 1-butanol, and ethyl ether (22) have been reduced to k_{ij} values, as summarized in Table IV. The 1974 predictions are supported to a large extent by the new information.

Table IV. Optimum k_{ij} Values for Polar–Nonpolar Binaries*

i	j	k_{ij}	t Range ($^{\circ}\text{C}$) (No. Points)
<i>i</i> = Ethanol			
	H ₂	0.16	25–75 (3)
	Ar	0.17	25–75 (3)
	Methane	0.15	25–75 (3)
	Ethylene	0.15	25 (1)
	Ethane	0.17	25–75 (3)
	N ₂ O	0.12	25–75 (3)
<i>i</i> = 1-Butanol			
	N ₂	0.24	25 (1)
	Ar	0.24	25 (1)
	Methane	0.20	25 (1)
	Ethane	0.17	25 (1)
<i>i</i> = Ethyl Ether			
	N ₂	0.22	25 (1)
	Ar	0.19	25 (1)
	Methane	0.12	25 (1)
	Ethane	0.06	25 (1)

* Ethanol binaries: Ref. 21. 1-Butanol and ethyl ether binaries: Ref. 22.

However, the more extensive data on the alcohol binaries (from Table IV and from Ref. 1) suggest certain trends in the k_{ij} values that warrant further examination.

The average k_{ij} value for the ten alcohol–hydrocarbon binaries is 0.16 ± 0.03 , which is in good agreement with the 1974 value. Indeed, if the eight alcohol–inorganic binaries also are included, the average becomes 0.15 ± 0.04 . However, it appears that k_{ij} increases in going from methanol to ethanol and then to 1-butanol.

The apparent trends in the k_{ij} values for the alcohol–nonpolar binaries are shown in Figure 5, where k_{ij} has been plotted vs. $V_{c,p}$, the critical volume of the nonpolar component. (Plotting k_{ij} vs. $V_{c,i}/V_{c,j}$ did not bring the data close together or even suggest more definite trends.) At a given $V_{c,p}$, k_{ij} definitely increases with the carbon number of the alcohol. The k_{ij} also increases as $V_{c,j}$ increases, at least for the methanol and ethanol binaries, although a limiting k_{ij} is approached at $V_{c,j} \geq 300$ cm³/gmol. This limiting value is about 0.22, close to the average for all four 1-butanol binaries (see Table IV and Figure 5).

Also plotted on Figure 5 are the k_{ij} 's for the water binaries. If a trend similar to that for methanol is assumed, the limiting k_{ij} value is about 0.40 (in agreement with the 1974 recommendation for water–hydrocarbon binaries).

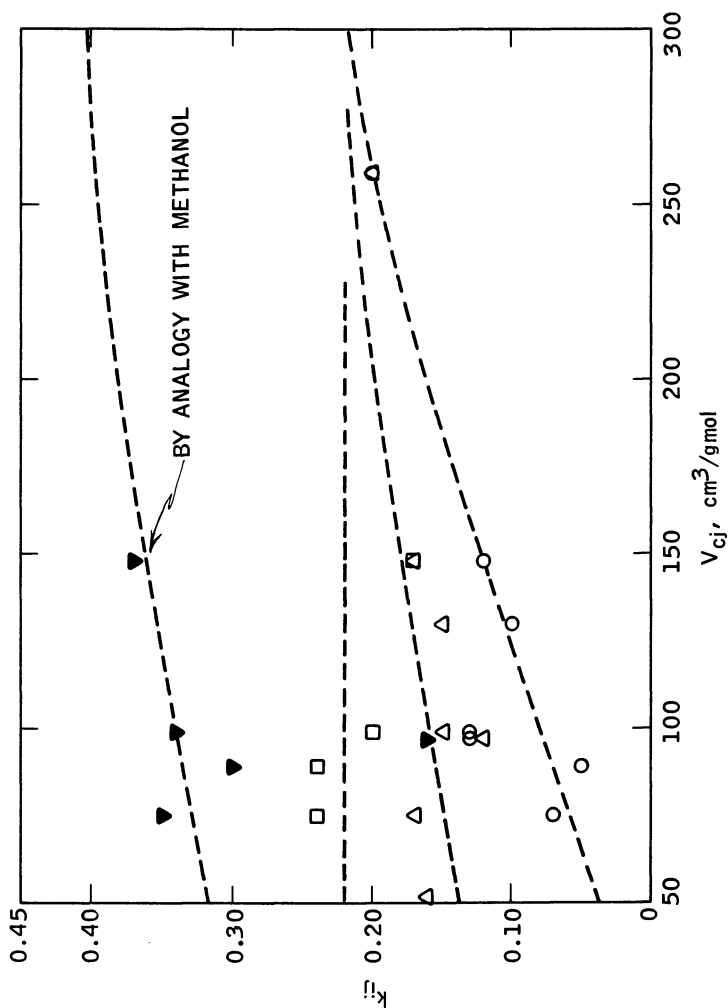


Figure 5. Optimum k_{ij} values for nonpolar binaries of alcohols and water: (O), methanol; (Δ), ethanol; ($\square$), 1-butanol; ($\blacktriangledown$), water.

The dashed lines in Figure 5 should not be given undue significance. They only present apparent trends and may prove useful in providing rough estimates for other alcohol-nonpolar or water-nonpolar binaries for which B_{ij} (and, hence, k_{ij}) data are not available. More information is needed to confirm and extend the usefulness of the trends indicated in Figure 5.

The more limited data for the hydrocarbon binaries of ethyl ether and ketones are relatively insensitive to $V_{c,j}$. In both cases, the present recommendations are essentially those made in 1974: 0.09 ± 0.02 for the ethyl ether-hydrocarbon and 0.13 ± 0.03 for the ketone-hydrocarbon binaries. However, it should be noted that the ethyl ether-inorganic binaries have a k_{ij} closer to 0.2 (*see* Table IV), while the k_{ij} for acetone-benzene is temperature-dependent because of complex formation (*see* Ref. 1).

CO₂ Binaries. The final group of binaries that will be examined is that of CO₂ with nonpolar and polar compounds. The thermodynamic properties of CO₂ differ significantly from those of similar nonpolar compounds, and this is the result of its very strong quadrupole moment. This effect is seen more clearly in the behavior of CO₂ in mixtures. King and his students have reported B_{ij} data for several CO₂ binaries; the resulting k_{ij} 's have been in Table V.

The k_{ij} values in Table V belong in two groups. The first one is the hydrocarbon and CCl₄ binaries, for which the k_{ij} is higher than what would be expected for the propane binaries. (Propane is the nonpolar homomorph of CO₂; the k_{ij} 's of the propane binaries can be estimated with Equation 15.) However, the k_{ij} 's for the second group, the binaries with the polar compounds, are significantly lower than those for the corresponding propane binaries.

Table V. Optimum k_{ij} Values for CO₂ Binaries

j	k_{ij}	t Range (°C) (No. Points)	References
Naphthalene	0.20 ^a	24–73 (9)	13
Phenanthrene	0.23	39–141 (7)	15
Anthracene	0.24	65–176 (6)	16
CCl ₄	0.20	25–75 (3)	23 ^b
Methanol	0.01	15–60 (5)	1, 24
Ethanol	0.07	25–75 (3)	21
1-Butanol	0.09	25 (1)	22
Ethyl ether	–0.11	25 (1)	22
Water	0.14	25–100 (4)	1, 25

^a B_{ij} data definitely suggest that k_{ij} is temperature-dependent (*see* text).

^b Optimum k_{ij} values have also been determined for the following nonpolar binaries of CCl₄: 0.22 (H₂); 0.16 (N₂); 0.15 (Ar); 0.09 (methane); 0.08 (ethylene).

The apparent increase in the k_{ij} values for the aromatic hydrocarbon and CCl_4 binaries is most probably caused by the effect of the quadrupole moment on the properties of CO_2 . That is, the strong quadrupole moment of CO_2 raises the values of all the k_{ij} 's for its nonpolar binaries. (The same is also true of the acetylene binaries.) However, the k_{ij} 's have a more complex behavior in the case of naphthalene- CO_2 , where a definite dependence of k_{ij} on temperature was found (13):

t ($^{\circ}\text{C}$)	k_{ij}
25	0.16
50	0.20
75	0.24

This is consistent with the observation of Najour and King (13) that "a relatively stable gas-phase, charge-transfer complex exists between naphthalene and CO_2 , with naphthalene being the donor and CO_2 the acceptor." The effect of the complex weakens as the temperature is increased. Apparently, the complex reaction must be weaker in binaries of CO_2 with the other hydrocarbons and CCl_4 because their optimum k_{ij} 's are temperature-independent.

The behavior of the CO_2 -polar binaries becomes much clearer when compared with the corresponding propane binaries. The k_{ij} 's for the two groups are as follows:

	CO_2	Propane (see Figure 5)	Difference
Methanol	0.01	0.16	-0.15
Ethanol	0.07	0.20	-0.13
1-Butanol	0.09	0.22	-0.13
Ethyl ether	-0.11	0.09 (average)	-0.20
Water	0.14	0.38	-0.24

The lowering of k_{ij} is essentially the same for all three alcohols, much larger for water, and surprisingly large for ethyl ether. As King and his students have pointed out, these effects are caused by specific chemical interaction between CO_2 and the polar molecules. However, the negative k_{ij} for CO_2 -ethyl ether is not easily explainable. Negative k_{ij} 's normally indicate the existence of a strong complex, probably formed by hydrogen bonding; an ether is incapable of forming a hydrogen bond with CO_2 .

Conclusions on Polar-Nonpolar Mixtures. The recommendations given in Ref. 1 for polar-hydrocarbon binaries are generally still valid. With the new k_{ij} 's reported here for alcohol-nonpolar binaries, however, it is possible to develop a correlation for the nonpolar binaries of water as well as for alcohols. This tentative correlation, which relates k_{ij} to V_{c_j} (j is the nonpolar component), is presented in Figure 5.

Finally, for the special case of the polar-CO₂ binaries, the k_{ij} 's are given by:

$$k_{ij} (\text{polar-CO}_2) = k_{ij} (\text{polar-propane}) - \Delta k_{ij}$$

For alcohols, $\Delta k_{ij} = 0.14 \pm 0.01$, while for ethers Δk_{ij} may be as high as 0.20.

Glossary of Symbols

Notation

- a = parameter in the VDW equation of state
- B = second virial coefficient
- $f^{(0)}, f^{(1)}$ = dimensionless terms of Equation 2
- I = first ionization potential
- k_{ij} = characteristic binary constant; *see* Equation 5
- n_c = carbon number
- P = pressure
- R = gas constant
- $t(T)$ = temperature (absolute)
- V = molar volume
- y_i = vapor mole fraction of Component i
- z = compressibility factor

Greek Letters

- ϵ = energy parameter
- σ = collision diameter
- ϕ_i = fugacity coefficient of Component i
- ω = acentric factor

Subscripts

- c = critical property
- i, j = property of Component i, j
- ij = characteristic property used in the calculation of the second virial cross-coefficient
- M = mixture property
- R = reduced property

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